

Boolean Matrices

In Chapter 6 we will be interested in matrices with only 0s and 1s as entries, called **Boolean matrices** (after George Boole, a nineteenth-century English mathematician; Boole also lent his name to *Boolean algebra*, which we will consider later in this book). Matrix **R** of Example 62 is a Boolean matrix. We can define an operation of Boolean matrix multiplication $\mathbf{A} \times \mathbf{B}$ on Boolean matrices using Boolean multiplication and Boolean addition instead of regular multiplication and addition. These are defined as follows:

Boolean multiplication: $x \wedge y = \min(x, y)$

Boolean addition: $x \vee y = \max(x, y)$

PRACTICE 56 Fill-in the following operation tables for Boolean multiplication and Boolean addition.

x	y	$x \wedge y$
1	1	
1	0	
0	1	
0	0	

x	y	$x \vee y$
1	1	
1	0	
0	1	
0	0	

Now take the tables from Practice 56 and substitute T for 1 and F for 0. They become the truth tables for conjunction and disjunction, respectively; for this reason, these operations are often called **Boolean and** (or **logical and**) and **Boolean or** (or **logical or**), which also explains the notation used for these operations. The operation of **Boolean matrix multiplication** $\mathbf{A} \times \mathbf{B}$ (on Boolean matrices of appropriate dimensions) is then defined by

$$c_{ij} = \bigvee_{k=1}^m (a_{ik} \wedge b_{kj})$$

We can also define two analogues of ordinary matrix addition (on Boolean matrices of the same dimensions): $\mathbf{A} \wedge \mathbf{B}$, where corresponding elements are combined using Boolean multiplication, and $\mathbf{A} \vee \mathbf{B}$, where corresponding elements are combined using Boolean addition.

EXAMPLE 75 Let **A** and **B** be Boolean matrices,

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Then

$$\mathbf{A} \wedge \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

and the Boolean product $\mathbf{A} \times \mathbf{B}$ is

$$\mathbf{A} \times \mathbf{B} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

PRACTICE 57 In Example 75, does $\mathbf{A} \times \mathbf{B} = \mathbf{A} \cdot \mathbf{B}$?

PRACTICE 58 In Example 75, compute $\mathbf{B} \times \mathbf{A}$.